Namrata Nadagouda

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OVERVIEW

- Research on developing methods for learning data efficient models based on active learning, human-in-the-loop learning and learning from human feedback/preferences
- Worked on applications such as preference learning, control parameter tuning, metric learning, and classification, and worked with images and sequential data
- Have a well rounded knowledge of ML theory and practice including both traditional ML and deep learning methods
- Experience with Python, MATLAB, PyTorch, SQL, Labelbox, Azure, AWS

EDUCATION

Ph.D. Electrical & Computer Engineering *December 2026 (Expected)*
Machine Learning and Digital Signal Processing
Georgia Institute of Technology, Atlanta, GA
Advisor: Prof. Mark Davenport
Thesis: "Building data efficient models using active learning and similarity comparisons"

M.S. Electrical & Computer Engineering December 2020
Machine Learning and Digital Signal Processing
Georgia Institute of Technology, Atlanta, GA

B.Tech. Electrical & Electronics Engineering May 2017
Digital Signal Processing and Digital System Design
National Institute of Technology Karnataka, Surathkal, India

PUBLICATIONS

N. Nadagouda, N. Ahad, M. Tucker and M. Davenport, "Active query synthesis for learning from preferences", *Under review.* Also presented at Women in Machine Learning (WiML) Symposium, co-located with *International Conference on Machine Learning (ICML), July 2026.*

N. Nadagouda, A. Xu and M. Davenport, "Active metric learning and classification using similarity queries", in *Uncertainty in Artificial Intelligence (UAI)*, August 2023. Also presented at Human in the Loop Learning Workshop, *Neural Information Processing Systems (NeurIPS), December 2022.*

A. McRae, A. Xu, J. Jin, **N. Nadagouda**, N. Ahad, P. Guan, S. Karnik and M. Davenport, "Delta Distancing: A Lifting Approach to Localizing Items From User Comparisons", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), May 2022.*

N. Nadagouda and M. Davenport, "Switched Hawkes Processes", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), June 2021.*

G. Canal, M. Connor, J. Jin, **N. Nadagouda**, M. O'Shaughnessy, C. Rozell and M. Davenport, "The Picasso Algorithm for Bayesian Localization Via Paired Comparisons in a Union of Subspaces Model", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), May 2020.*

ABSTRACTS

N. Ahad, **N. Nadagouda**, E. Dyer and M. Davenport, "Active learning for time instant classification", at Data-centric Machine Learning Research Workshop, *International Conference on Machine Learning (ICML), July 2023.*

Y. Teng, A. Mamuye, E. Mo, K. Zhu, R. Walker, **N. Nadagouda** and M. Davenport, "Range-Only Simultaneous Localization and Mapping using Paired Comparisons", at *IEEE Annual Conf. on RFID, April 2021.*

N. Nadagouda and M. Davenport, "Switched Hawkes Processes", at *Workshop on Recent Developments on Mathematical/Statistical approaches in Data Science (MSDAS), June 2019.*

ARTICLES

N. Nadagouda, "Journey of a researcher: Finding pleasure in the pathless woods", *American Ceramic Society Bulletin*, Student Perspectives, June/July 2020.

WORK EXPERIENCE

Applied Scientist Intern Summer 2024
ML Science Team, Samsara, San Francisco, CA

- Worked on active learning based traffic sign detection model using trip still images from dashcams
- Built the infrastructure to actively select images for labeling using a pre-trained distilled model and fine-tune the model using the labels obtained
- Also, involved data preparation which comprised of text prompt based collection of data using a CLIP based similarity search tool and labeling using Labelbox
- Conducted initial experiments with varying batch sizes of acquired data and the findings were handed off to the team for further investigation

Intern Summer 2018
Hedge Fund Start-up, Atlanta, GA

- Worked on data management and data pre-processing of stock trade data stored in SQL databases.
- Also, involved performance evaluation of trading algorithms on Microsoft Azure platform

Research Intern, Microarchitecture Research Lab Fall 2016
Intel India - Intel Labs, Bangalore, India

Research Intern Summer 2016
Department of Electrical Communication Engineering
Indian Institute of Science, Bangalore, India

TALKS

Active query synthesis for preference learning 2026
CSIP Seminar, Georgia Tech, Atlanta, GA

Active query synthesis for preference learning 2025
ML@GT Student Conference, Georgia Tech, Atlanta, GA
AWM Workshop at SIAM Annual Meeting, Montreal, Canada

Active learning for traffic sign detection 2024
Samsara, San Francisco, CA

	Building data efficient models using active learning and similarity comparisons *Thesis presentation at Georgia Tech, Atlanta, GA*	2022
	Switched Hawkes Processes *Student Seminar at Georgia Tech, Atlanta, GA*	2019
AWARDS	2nd place, Presentations *ML@GT Student Conference* Georgia Tech	September 2025
Registration and Travel Awards	AWM Workshop at SIAM Annual Meeting *Travel Funding*	July 2025
	NeurIPS *Conference Volunteer Financial Assistance* *WiML Travel Funding*	December 2023
	Women in Data Science and Mathematics Institute for Pure and Applied Mathematics *Workshop Financial Assistance* *GT Career Development Funding*	August 2023
	UAI *Conference Funding*	August 2023
	ICML *GT SGA Conference Funding* *CoE Professional Development Funding* *WiML Funding*	August 2023
	NeurIPS *GT SGA Conference Funding* *CoE Professional Development Funding*	December 2022
	Deep Learning Theory Workshop and Summer School Simons Institute *Workshop Financial Assistance*	August 2022
	The Mathematics of Machine Learning Women and Math Program Institute for Advanced Study *Workshop Financial Assistance*	May 2022
	Recent Developments on Mathematical/Statistical approaches in Data Science University of Texas, Dallas *Workshop Financial Assistance*	June 2019
Hackathon	Winner, Technical Track, Hacklytics Data Science at Georgia Tech	February 2019
Academic Awards and Scholarships	NITK Institute Gold Medal 1986 Batch Gold Medal	2017 2017

Prof. M. R. Shenoy Memorial Prize 2017
Prof. K. M. Hebbar Gold Medal 2017
NITK Surathkal Merit Scholarship 2013 - 2017

SERVICE

Reviewer, ICLR 2027
Reviewer, NeurIPS 2026
Reviewer, WiML Workshop at NeurIPS 2025
Judge, AWM Poster Session at SIAM Annual Meeting 2025
Mentor, AWM Workshop at SIAM Annual Meeting 2025
Reviewer, Workshop Proposals for NeurIPS 2025
Reviewer, WiML Workshop at NeurIPS 2024
Reviewer/Volunteer, WiML workshop at NeurIPS 2023
Volunteer, NeurIPS Conference 2023
Volunteer, UAI Conference 2023
Volunteer, WiML Workshop at ICML 2023
Member, GT Mural Team 2022
Reviewer, AIStats Conference 2021
Reviewer, GT President's Undergraduate Research Award 2021 - 2022
Panelist, GT ECE ORS Graduate Panel 2020
Volunteer, GT ECE Prospective PhD student visit 2019 - 2022
Teaching Volunteer, Shiksha, ACM NITK Student Chapter 2016
Student Representative, NITK Student Council 2013 - 2014

TEACHING EXPERIENCE

Mentor for undergraduate students
Yue Teng 2020 - 2022
Amran Mamuye, Eunsan Mo, Kerui Zhu, Robert Walker 2020 - 2021
Guided the above students to work on a research project focused on simultaneous localization and mapping using paired comparisons of distances.

Graduate Teaching Assistant Spring 2021
Georgia Tech ECE 6270 - Convex Optimization

Teaching Assistant June 2019
Hands-on-Tech Georgia Tech Day Camp - Machine Learning

Graduate Teaching Assistant Spring 2019
Georgia Tech CS 4641 - Machine Learning

Graduate Teaching Assistant Fall 2018
Georgia Tech ECE 8843/ISYE 8843/CS 8803/BMED 8813 -
Mathematical Foundations of Machine Learning

RESEARCH EXPERIENCE

Human Color Estimation Study Fall 2026
A user study was conducted where each participant was shown a target color and answered 20 sequential queries of the form 'Which among these two colors is closer to the target?'. We compare the performance of *confidence aware* response model against other response models. The estimation error, cross-entropy and accuracy of each model's predictions on the participant's choices, and mean self-reported confidence, averaged over all the participants, were compared. Our *confidence aware* model is the best on all four, and its queries receive the highest confidence, thereby proving it selects queries that balance both informativeness and response confidence.

Control parameter tuning using active synthesis Fall 2025 - present
Active preference learning was applied to optimize subjective controller gains for

mobile robots. The framework dynamically synthesizes continuous-space queries to compare different trajectory parameters, asking users which control behavior they prefer. By translating this qualitative human feedback into a structured search, the method iteratively identifies optimal control parameters with minimal queries.

Active query synthesis for preference learning Fall 2022 - present
Proposed a *confidence aware* response model for pairwise comparisons, in which response reliability depends on the similarity of the compared options, to better capture human choices and improve preference learning. To avoid the computational cost of pool-based active learning, the *Info-Synth* query synthesis framework directly generates informative queries by maximizing mutual information in closed form, up to a one-dimensional search. For fixed datasets, the method employs tailored approximation strategies to select the closest available pairs to these synthesized queries, matching the sample efficiency of pool-based methods at lower computational cost.

Active learning for time series data Summer 2023
Developed a method for classifying individual instants of time series data. The data consists of features which repeat at regular intervals and the existence of these correlations poses a unique challenge for active label selection.

Unified framework for active learning Fall 2019 - Spring 2023
Developed a unified query framework for active learning based on nearest neighbor queries. This method can be applied to any problem which involves learning a representation of the dataset that reflects the underlying similarity. Demonstrated the performance of the method for active metric learning and active image classification using deep neural networks.

Active similarity learning and manifold graphs Fall 2019 - Spring 2023
Implemented active image classification strategies for semi-supervised classification on CIFAR-100 and DomainNet datasets. This project was funded by the *DARPA LwLL – Learning with Less labels* program.

Preference learning Fall 2019 - Summer 2022
The problem consists of estimating a user's preferences over a set of items. We use the *ideal point* model to localize a user in an embedding of items. Worked on a variety of problems involving localizing new items and users.

Switched Hawkes Processes Fall 2018 - Summer 2019
Developed the Switched Hawkes Process which can be used to model systems in which the parameters of the process dynamically change depending on some (known) external state. We propose a simple maximum likelihood estimation approach and apply our model to a real-world traffic sensor dataset to study traffic patterns during different configurations of the traffic lights at an intersection.

WORKSHOPS ATTENDED

Women in Data Science and Mathematics — August 2023
Institute for Pure and Applied Mathematics
University of California, Los Angeles

Deep Learning Theory Workshop and Summer School — August 2022
Simons Institute, University of California, Berkeley

The Mathematics of Machine Learning — May 2022
Women and Math Program
Institute for Advanced Study, Princeton, NJ

Algorithmic Learning Theory Mentoring Workshop — March 2022
Online

Foundation of Data Science Summer School — August 2019
Georgia Institute of Technology, Atlanta, GA

Recent Developments on Mathematical/Statistical approaches in Data Science — June 2019
University of Texas at Dallas, Richardson, TX